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Air vent

Abstract

An air vent includes a duct including an inlet and an outlet, a mesh cover including a plurality of mesh holes communicating with the outlet of the duct, a wing provided in the duct and configured to laterally adjust a wind direction, air dampers which move forward and backward directions in the duct, wherein the air damper opens and closes the gap between the duct outlet and the mesh cover.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of U.S. application Ser. No. 17/574,870 filed on Jan. 13, 2022, which claims the benefit under 35 USC § 119(a) of Korean Patent Application No. 10-2021-0007079, filed on Jan. 18, 2021 in the Korean Intellectual Property Office. The entire disclosures of U.S. application Ser. No. 17/574,870 and Korean Patent Application No. 10-2021-0007079 are incorporated herein by reference for all purposes.

BACKGROUND

1. Field of the Invention

(1) The present invention relates to an air vent.

2. Discussion of Related Art

(2) In the recent vehicle market, many efforts to slim a structure of an air vent are being made in order to apply a next generation slim cockpit design, and as part of the efforts, various attempts are being made in consideration of design and function.

(3) Particularly, in the next generation cockpit design, since it is not preferred for a wing of the air vent, which is still used the most, to be exposed to the outside, providing a knob, with which a wind direction of the wing may be adjusted manually, has also become virtually unnecessary.

(4) Even without considering the design, there are many cases in which wind directly comes into contact with a passenger's body and causes inconvenience when a wing type air vent, whose wind direction is directly adjusted by a user, is used. Although there was a precedent in which a mesh type product with a plurality of holes formed to avoid the direct wind was released, in the structure in which the wind simply flows through mesh holes, since some problems, in that air resistance increases and there is no wind orientation, may be caused, it was difficult to satisfy a user's need to manipulate the wind.

SUMMARY OF THE INVENTION

(5) The present invention is directed to providing an air vent for a vehicle, in which a mesh cover is applied to a front portion (duct outlet side) of an air vent to reduce a surrounding pressure so as to adjust a wind direction of windless (indirect) wind and which induces the occurrence of a Coanda effect using a skin surface to satisfy the needs of manipulating the wind direction of a user and allows a product upgrading strategy to be established.

(6) In addition, the present invention is directed to providing a technology of adjusting opening and closing of a gap between an outlet of a duct and a mesh cover using an air damper operating in conjunction with a link joint structure so that indirect wind flows in a direction desired by a user.

(7) Objectives to be achieved by the present invention are not limited to the above-described objectives, and other objectives, which are not described above will be clearly understood by those skilled in the art from the following descriptions.

(8) According to an aspect of the present invention, an air vent includes a duct including an inlet and an outlet, a mesh cover including a plurality of mesh holes communicating with the outlet of the duct, a wing provided in the duct and configured to laterally adjust a wind direction, air dampers which are disposed on upper and lower ends of the wing in the duct and selectively open or close gaps between the outlet of the duct and upper and lower ends of the mesh cover, and a link part which moves the air dampers in the duct in forward and backward directions.

(9) The upper and lower ends of the mesh cover may be spaced apart from the outlet of the duct to allow air introduced through the inlet of the duct to flow along a garnish extending from the outlet of the duct in upward and downward directions and to be discharged through the outlet.

(10) The link part may include an upper arm connected to the air damper disposed on the upper end of the wing, a lower arm connected to the air damper disposed on the lower end of the wing, a connecting rod disposed between and connected to the upper arm and the lower arm, and link pins which connect an end portion of the upper arm, an end portion of the lower arm, and end portions of the connecting rod which are in contact with each other.

(11) The connecting rod may include a link joint structure divided into a plurality of rods.

(12) One end of the connecting rod may be fixed to the other end of the upper arm, and the other end of the connecting rod may be fixed to one end of the lower arm.

(13) The upper arm and the lower arm may be provided with the connecting rod interposed therebetween and may be perpendicular to the connecting rod in different directions.

(14) The link pins connected to the upper arm and the lower arm may be movable in link holes disposed in the upper and lower ends of the wing in the forward and backward directions. The mesh holes of the mesh cover may be disposed in a circular or diagonal shape in a mesh plate of the mesh cover.

- (15) The air vent may be configured to be disposed inside a dashboard in a vehicle, and configured to receive air from an air conditioner of the vehicle through the inlet and to discharge the air into the vehicle through the outlet.
- (16) The mesh cover may include a material the same as a material of a garnish of a board, which is adjacent to the outlet of the duct, in the vehicle.
- (17) According to another aspect of the present invention, there is provided an air vent for a vehicle including a duct including an inlet and an outlet, a mesh cover communicating with the outlet of the duct, a wing provided in the duct and configured to laterally adjust a wind direction, air dampers disposed on upper and lower ends of the wing in the duct, and a driving actuator configured to move the air dampers in the duct in forward and backward directions.
- (18) The air dampers may be disposed above and under the wing and may selectively open or close gaps between the duct and the mesh cover to guide air introduced through the inlet of the duct in one direction toward an upper or lower end of the mesh cover.
- (19) The air damper may include an upper damper positioned at an upper end in the duct and a lower damper positioned at a lower end in the duct, wherein a connecting link divided by a rotation shaft may be provided between the upper damper and the lower damper.
- (20) The connecting link may include an upper connecting bar hinge-connected to the upper damper and a lower connecting bar hinge-connected to the lower damper.
- (21) In this case, the upper connecting bar and the lower connecting bar may be hinge-connected by the rotation shaft, and the rotation shaft may be rotatably fixed to a designated section in the duct.
- (22) The inlet may be configured to communicate with an air conditioner of a vehicle and the outlet may be configured to direct the air into the vehicle.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other objects, features and advantages of the present invention will become more apparent to those of ordinary skill in the art by describing exemplary embodiments thereof in detail with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a schematic view illustrating an air vent for a vehicle according to a first embodiment of the present invention;
- (3) FIGS. 2 to 4 are views illustrating a state in which a wind direction of indirect wind is adjusted by forward and backward movement of an air damper in the air vent for a vehicle according to the first embodiment of the present invention;
- (4) FIGS. 5 to 7 are views illustrating a state in which a wind direction of indirect wind flowing through a mesh cover is adjusted by lateral adjustment of a wing in the air vent for a vehicle according to the first embodiment of the present invention;
- (5) FIG. 8 is a schematic view illustrating a air vent for a vehicle according to a second embodiment of the present invention; and
- (6) FIGS. 9 to 11 are views illustrating a state in which a wind direction of indirect wind is adjusted by forward and backward movement of an air damper in the air vent for a vehicle according to the second embodiment of the present invention.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

- (7) Advantages and features of the present invention and methods of achieving the same will be clear with reference to the accompanying drawings and the following detailed embodiments. However, the present invention is not limited to the embodiments to be disclosed below but may be implemented in various different forms, the embodiments are provided in order to fully explain the present invention and fully explain the scope of the present invention for those skilled in the art,

and the scope of the present invention is defined by the appended claims. Meanwhile, the terms used herein are provided to describe embodiments of the present invention and not for purposes of limitation. In the specification, unless the context clearly indicates otherwise, the singular forms include the plural forms. The terms “comprise” or “comprising” used in the specification specify some stated components, steps, operations and/or elements but do not preclude the presence or addition of one or more other components, steps, operations and/or elements.

(8) Hereinafter, exemplary embodiments of the present invention will be described with reference to the accompanying drawings.

First Embodiment

(9) FIG. 1 is a schematic view illustrating an air vent for a vehicle according to one embodiment of the present invention.

(10) Referring to FIG. 1, an air vent **100** for a vehicle according to one embodiment of the present invention is installed inside a dashboard in a vehicle and changes a mode of air introduced from a vehicle air conditioner (not shown) to a direct wind mode or indirect wind (windless) mode to guide the air into the vehicle. In this case, the direct wind mode is a setting implemented to allow wind to directly flow to a user, and the indirect wind (windless) mode is a setting implemented to block wind in advance for a user who wants to avoid direct wind.

(11) Generally, when the direct wind mode is used, it is absolutely necessary to avoid lateral wind direction interference due to a monitor (not shown), and when the indirect wind mode is used, since the wind direction interference avoidance is not necessary, a cover is just closed.

(12) However, a first embodiment of the present invention proposes a mechanical mechanism capable of internally adjusting a wind direction for wind directivity.

(13) To this end, an air vent **100** for a vehicle according to the first embodiment of the present invention includes a duct **110**, a mesh cover **120**, a wing **130**, a link part **140**, air dampers **150**, and a driving actuator **160**.

(14) The duct **110** includes an inlet communicating with a vehicle air conditioner, an outlet directed into the vehicle, and an air path disposed between the inlet and the outlet.

(15) The mesh cover **120** partially closes the outlet of the duct **110**.

(16) The mesh cover **120** includes a mesh plate **121** forming a body and mesh holes **122** formed to be spaced apart from each other in the mesh plate **121**. The mesh hole **122** may communicate with the outlet of the duct **110** and may be formed as a plurality of mesh holes **122**.

(17) A direction of wind discharged into the vehicle may also be adjusted according to a layout of the mesh holes **122**. That is, the mesh holes **122** may also be formed to be inclined in one direction in the mesh plate **121** to guide the direction of the wind. For example, the mesh holes **122** may be disposed in a circular or diagonal shape in the mesh plate **121**.

(18) Upper and lower ends of the mesh cover **120** are disposed to be spaced apart from the outlet of the duct **110**, and air introduced from the inlet of the duct **110** flows along a garnish **101** extending from the outlet of the duct **110** in upward and downward directions and is discharged into the vehicle.

(19) In this case, since the mesh cover **120** is formed of a material which is the same as a material of the garnish **101**, which is close to the outlet of the duct **110**, of a board in the vehicle, an exterior design can be improved, and a Coanda effect can be effectively induced.

(20) The wing **130** may be provided in the duct **110** and may laterally adjust a wind direction of air discharged through the outlet of the duct **110**.

(21) The air dampers **150** are disposed in the duct **110**. The air dampers **150** are disposed on upper and lower ends of the wing **130** and selectively open or close gaps between the outlet of the duct **110** and the upper and lower ends of the mesh cover **120**.

(22) The link part **140** has a structure which moves the air dampers **150** in the duct **110** in forward and backward directions. In this case, the forward direction means a direction toward the inlet of the duct **110**, and the backward direction means a direction toward the outlet of the duct **110**.

(23) The link part **140** includes an upper arm **141**, a lower arm **142**, a connecting rod **143**, and link pins **145**.

(24) The upper arm **141** is connected to the air damper **150** disposed on the upper end of the wing **130**.

(25) The lower arm **142** is connected to the air damper **150** disposed on the lower end of the wing **130**.

(26) The connecting rod **143** is disposed between and connected to the upper arm **141** and the lower arm **142**. The connecting rod **143** may be formed to have a link joint structure divided into a plurality of rods. For example, one end of the connecting rod **143** is fixed to the other end of the upper arm **141**, and the other end of the connecting rod **143** is connected to one end of the lower arm **142**.

(27) In this case, the upper arm **141** and the lower arm **142** may be disposed with the connecting rod **143** disposed therebetween and may be perpendicular to the connecting rod **143** in different directions. For example, the upper arm **141**, the connecting rod **143**, and the lower arm **142** may be connected in a “custom character” shape.

(28) The link pins **145** connect an end portion of the upper arm **141**, an end portion of the lower arm **142**, and end portions of the connecting rod **143** which are in contact with each other. The link pins **145** may move in link holes **144** disposed in the upper and lower ends in the forward and backward directions.

(29) FIGS. **2** to **4** are views illustrating a state in which a wind direction of indirect wind is adjusted by forward and backward movement of the air damper in the air vent for a vehicle according to the first embodiment of the present invention.

(30) Referring to FIGS. **2** to **4**, in the structure of the air vent **100** for a vehicle, a phenomenon may occur in which a pressure in a wind region, in which a wind speed is high, decreases and wind is collected in the region.

(31) A region from the outlet of the duct **110** to an upper end or a lower end of an end portion of the mesh cover **120** has a structural feature in which high speed wind is generated to allow low speed windless wind to follow the high-speed wind. In such a structure of the mesh cover **120**, a Coanda effect may be generated along a skin surface of the garnish **101**. This means that wind direction performance is improved.

(32) When the air damper **150** moves in the duct **110** in the forward and backward directions, a gap between the outlet of the duct **110** and the mesh cover **120** is opened or closed. That is, due to forward and backward movement of the air damper **150**, in the indirect wind (windless) mode, directivity, which is a disadvantage of the mesh cover **120**, may be generated.

(33) The air vent **100** for a vehicle in FIG. **2** has the structure in which air flowing from the inlet of the duct **110** to the outlet thereof may flow into the vehicle through the mesh holes **122** of the mesh cover **120** and gaps above and under the mesh cover **120**. Through this, the air discharged through the outlet of the duct **110** is diffused.

(34) In the air vent **100** for a vehicle in FIG. **3**, since the air damper **150** closes a gap between a lower end portion of the outlet of the duct **110** and a lower end portion of the mesh cover **120**, air flowing from the inlet of the duct **110** toward the outlet thereof is discharged in the upward direction.

(35) In the air vent **100** for a vehicle in FIG. **4**, since the air damper **150** closes a gap between an upper end portion of the outlet of the duct **110** and an upper end portion of the mesh cover **120**, air flowing from the inlet of the duct **110** toward the outlet thereof is discharged in the downward direction.

(36) FIGS. **5** to **7** are views illustrating a state in which a wind direction of indirect wind flowing through the mesh cover is adjusted by lateral adjustment of the wing in the air vent for a vehicle according to the first embodiment of the present invention.

(37) Referring to FIGS. **5** to **7**, basically, since the mesh cover **120** partially closes the outlet of the

duct **110**, the indirect wind mode, which is similar to the windless mode, of the air vent **100** for a vehicle may be a basic mode thereof.

(38) However, when the mesh cover **120** is close to the wing **130** and an angle of the wing **130** is electrically changed by the driving actuator **160** (see FIG. **1**) to change a wind direction, the direct wind mode may be applied through the mesh holes **122** of the mesh cover **120**.

(39) Of course, since it is problematic for wind to flow to the user through the mesh holes **122** in the direct wind mode, as described above, the gap between the outlet of the duct **110** and the mesh cover **120** may be adjusted according to driving of the air damper **150** (see FIG. **1**) so that the wind may flow in a direction desired by the user. In this case, an important point is a set distance between the mesh cover **120** and the wing **130**. When the set distance is greater than a set reference distance, air flowing into the vehicle through the mesh cover **120** is not affected by the wing **130**.

(40) In this case, since the wing **130** close to the mesh cover **120** affects a wind direction due to the mesh cover **120**, when the mesh cover **120**, to which a leftward wing having a size of about 3 mm is coupled, is applied to a front portion of the wing **130**, a directivity angle of the wing **130** faces a right side, and air flowing into the vehicle through the mesh cover **120** may be adjusted to a leftward direction.

Second Embodiment

(41) FIG. **8** is a schematic view illustrating an air vent for a vehicle according to a second embodiment of the present invention;

(42) An air vent **200** for a vehicle illustrated in FIG. **8** includes a duct **210**, a mesh cover **220**, a wing **230**, air dampers **250**, a driving actuator **260**, and a connecting link **270**.

(43) Regarding the above configuration, contents overlapping with the first embodiment will be omitted, and in the second embodiment, structural features different from those of the first embodiment will be described.

(44) The air damper **250** includes an upper damper **251** positioned at an upper end in the duct **210** and a lower damper **252** positioned at a lower end in the duct **210**.

(45) The connecting link **270** divided by a rotation shaft **273** is positioned between the upper damper **251** and the lower damper **252**.

(46) The connecting link **270** includes an upper connecting bar **271**, a lower connecting bar **272**, the rotation shaft **273**, link holes **274**, and link pins **275**.

(47) The upper connecting bar **271** is hinge-connected to the upper damper **251**.

(48) The lower connecting bar **272** is hinge-connected to the lower damper **252**.

(49) The upper connecting bar **271** and the lower connecting bar **272** are hinge-connected by the rotation shaft **273**. The rotation shaft **273** is formed in a structure rotatably fixed to a designated section in the duct **210**. In this case, the designated section means a position at which the upper connecting bar **271** and the lower connecting bar **272** are rotatable about a position at which the rotation shaft **273** is fixed. In order to adjust a wind direction of air, it is advantageous for a rotation section of the rotation shaft **273** to be positioned in a driving radius of the wing **230** at an outlet side of the duct **210**.

(50) FIGS. **9** to **11** are views illustrating a state in which a wind direction of indirect wind is adjusted by forward and backward movement of the air damper in the air vent for a vehicle according to the second embodiment of the present invention.

(51) Referring to FIGS. **9** to **11**, a wind direction adjustment method of the air vent **200** for a vehicle according to the second embodiment of the present invention is basically similar to that of the first embodiment.

(52) However, as illustrated in FIG. **9**, a windless mode is different in that the upper damper **251** and the lower damper **252** of the air damper **250** completely close gaps between the outlet of the duct **210** and upper and lower ends of the mesh cover **220**.

(53) When the connecting link **270** rotates about the rotation shaft **273** and the upper connecting bar **271** and the lower connecting bar **272** move in forward and backward directions, the upper

damper 251 and the lower damper 252 operating in conjunction therewith may move, and a direction of air flowing into the vehicle through gaps generated due to the movement and the mesh holes 222 may be determined.

(54) In an air vent for a vehicle according to the present invention, since a mesh cover is applied to an outlet side of a duct through which wind is introduced into a vehicle, the marketability and functionality of a product can be improved, and thus an upgrading strategy of a next generation cockpit design can be established. Particularly, since the air vent for a vehicle is formed in a structure capable of inducing a Coanda effect using a skin surface, wind directivity can be improved.

(55) In addition, in the air vent for a vehicle, since opening and closing of a gap between an outlet of the duct and the mesh cover is adjusted using an air damper operating in conjunction with a link joint structure, indirect wind flows in a direction desired by a user so that needs of manipulating a wind direction of the user can be satisfied.

(56) The present invention is not limited to the above-described embodiments and may be variously modified and implemented within a range allowed by the technical spirit of the present invention.

Claims

1. An air vent, comprising: a duct including an inlet and an outlet; a mesh cover including a plurality of mesh holes communicating with the outlet of the duct; a wing provided in the duct and configured to laterally adjust a wind direction; and air dampers which move forward and backward directions in the duct; wherein each air damper opens and closes a gap between the duct outlet and the mesh cover.
2. The air vent of claim 1, wherein the mesh cover includes a mesh plate forming a body and mesh holes spaced apart from each other in the mesh plate.
3. The air vent of claim 2, wherein the mesh holes is inclined in one direction in the mesh plate to guide the wind direction.
4. The air vent of claim 1, wherein the mesh cover is formed of a material which is the same as a material of a garnish, which is close to the outlet of the duct, of a board in a vehicle.
5. The air vent of claim 1, wherein a link part which moves the air dampers in the duct in the forward and backward directions.
6. The air vent of claim 1, wherein the link part includes: an upper arm connected to one of the air dampers disposed on an upper end of the wing; a lower arm connected to another of the air dampers disposed on a lower end of the wing; a connecting rod disposed between and connected to the upper arm and the lower arm; and link pins which connect an end portion of the upper arm, an end portion of the lower arm, and end portions of the connecting rod which are in contact with each other.
7. The air vent of claim 1, wherein the air dampers are disposed above and under the wing and selectively open or close gaps between the duct and the mesh cover to guide air introduced through the inlet of the duct in one direction toward an upper or lower end of the mesh cover.
8. An air vent, comprising: a duct including an inlet and an outlet; a mesh cover communicating with the outlet of the duct; a wing provided in the duct and configured to laterally adjust a wind direction; air dampers disposed on upper and lower ends of the wing in the duct; and a driving actuator configured to move the air dampers in the duct in forward and backward directions, wherein the air dampers include: an upper damper positioned at an upper end in the duct; and a lower damper positioned at a lower end in the duct, wherein a connecting link divided by a rotation shaft is disposed between the upper damper and the lower damper.
9. The air vent of claim 8, wherein the connecting link includes: an upper connecting bar hinge-connected to the upper damper; and a lower connecting bar hinge-connected to the lower damper, wherein the upper connecting bar and the lower connecting bar are hinge-connected by the rotation

shaft, and the rotation shaft is rotatably fixed to a designated section in the duct.

10. The air vent of claim 9, wherein the designated section of the rotation shaft is positioned in a driving radius of the wing in the duct outlet.

11. The air vent of claim 8, wherein the upper and lower ends of the mesh cover are spaced apart from the outlet of the duct to allow air introduced through the inlet of the duct to flow along a garnish extending from the outlet of the duct in the upward and downward directions and to be discharged through the outlet.
